

Credit Risk Assessment Pipeline

Data Lake Information – RAW Data in ADLS Gen2

Scope: This document covers only the data lake information required to store the source datasets as RAW data in the Azure Data Lake Storage Gen2 (ADLS Gen2) storage account after the data model is ready..

1. Objective

Design the storage structure and supporting information required to land and preserve the original Credit Risk Dataset CSV files in ADLS Gen2 RAW storage. The RAW layer acts as the source-of-truth copy of the data received before downstream processing.

2. Source Dataset Information

Attribute	Details
Source	Kaggle – Credit Risk Dataset
Source Format	CSV
Data Type	Structured tabular data
Example Data	Applicant details, loan amount, income, credit score, employment/credit history and related credit attributes
Destination	Azure Data Lake Storage Gen2
Data Layer	RAW
Ingestion Pattern	Daily/periodic batch source file landing

3. ADLS Gen2 Storage Account

Create an Azure Storage Account with Hierarchical Namespace enabled so that it can be used as ADLS Gen2.

Example logical storage account name: creditriskstg

The actual storage account name should follow the organization's Azure naming convention and Azure naming requirements.

4. RAW Container / Filesystem

Create a dedicated container/filesystem named raw for the original source datasets.

Logical structure:

ADLS Gen2 Storage Account



raw
↓
source datasets

5. RAW Folder Structure

Recommended folder hierarchy:

```
/raw/  
  credit_data/  
    YYYY/  
      MM/  
        DD/  
          credit_risk_dataset.csv
```

Example:

```
/raw/credit_data/2026/08/27/credit_risk_dataset.csv
```

The Year/Month/Day structure supports daily batch processing, historical tracking, reprocessing of a specific batch, and easier source-data management.

6. RAW Data Storage Rules

The RAW layer should preserve the source file as received. The purpose is to retain an original copy for traceability, audit and possible reprocessing.

- Do not remove duplicate records in the RAW file.
- Do not fill or replace NULL values.
- Do not calculate risk scores.
- Do not apply business transformations.
- Do not change the business meaning of source columns.
- Do not overwrite historical source batches without an approved requirement.

7. File Naming Convention

Use a consistent naming convention so files can be identified by source and ingestion date.

Recommended example:

```
credit_risk_2026_08_27.csv
```

If the source contains multiple datasets, use names such as:

```
applicant_2026_08_27.csv
```

```
loan_2026_08_27.csv
```

```
credit_history_2026_08_27.csv
```

8. Data Lake Metadata

Maintain metadata for each landed source file:

- Source system / source name
- Dataset or file name
- File format
- File arrival timestamp
- Ingestion date
- File size
- Record count
- Processing status
- Source path
- Target RAW path

9. Source-to-RAW Mapping

Source	ADLS Gen2 RAW Location	Format
Kaggle Credit Risk Dataset	/raw/credit_data/YYYY/MM/DD/	CSV

10. Access and Security

Access to the ADLS Gen2 RAW layer should be controlled using Microsoft Entra ID and Azure RBAC/ACLs.

- Define read permissions for authorized data engineers and processing services.
- Define write permissions for the approved ingestion process.
- Restrict delete or modification permissions.
- Apply least-privilege access.
- Protect credit-risk data from unrestricted access.

11. Retention Requirement

RAW source files should be retained according to the project's business, audit and compliance requirements. If an exact retention period has not been approved, record it as a requirement to be finalized rather than assuming a specific number of days.

12. ADLS Gen2 RAW Storage Diagram

Kaggle Credit Risk CSV



Azure Data Lake Storage Gen2



Storage Account: creditriskstg

↓
Container: raw
↓
credit_data/YYYY/MM/DD/
↓
credit_risk_dataset.csv

13. Deliverables for This P2 Point

- ☐ Source dataset information
- ☐ ADLS Gen2 storage account information
- ☐ RAW container/filesystem information
- ☐ RAW folder hierarchy
- ☐ File naming convention
- ☐ Metadata requirements
- ☐ Access and security requirements
- ☐ Retention requirement
- ☐ Source-to-RAW mapping
- ☐ RAW storage architecture diagram

14. Completion Criteria

This P2 point can be considered complete when the team has documented where the source CSV files will be stored in ADLS Gen2, the RAW folder/path convention, naming standard, metadata, access controls and retention requirements, and has verified that the intended source files can be landed in the RAW location.